

# **Flow through a 90-degree bend**

## Abstract

This report aims to identify and explain the flow of a fluid through a 90-degree pipe bend modelled using STARCCM++ (a CFD tool). Two types of models are considered, 2D pipe flow and 3D pipe flow, both using a fine mesh, where the mesh gets finer nearer the walls of the models, to more accurately provide the data of how the flow interacts with the boundary. The key aspects considered when modelling these pipe sections were to quantify the pressure drop between the inlet and outlet for turbulent flows of different Reynold's number. These models are qualitatively analysed explaining the inner workings of the flow behaviour and why certain characteristics are present within the pressure and velocity fields by linking the findings to theory and literature. Furthermore, the obtained pressure drops from the simulation are then compared to analytically obtained values of pressure drop to ascertain any discrepancies between an ideal model and a simulated model. The key findings regarding this that there are large discrepancies between the 2D simulations and empirical models, ranging to a maximum variation of 387.4%, whereas the 3D show a closer correlation between the simulation and empirical models, with a variation of only up to 37.4%. This represents the closeness of the CFD model to expectations, showing that the more complex and realistic the model, the closer the values are to the empirical estimates.

## Contents

|     |                                                     |    |
|-----|-----------------------------------------------------|----|
| 1   | Introduction.....                                   | 1  |
| 2   | 2-D Results.....                                    | 1  |
| 2.1 | How the pipe was modelled .....                     | 1  |
| 2.2 | 2-D CFD velocity and pressure fields analysis ..... | 2  |
| 2.3 | Remarks on graphs of results .....                  | 3  |
| 3   | 3-D Results.....                                    | 3  |
| 4   | Analysis of pressure drop data .....                | 4  |
| 4.1 | 2D pressure drop analysis .....                     | 4  |
| 4.2 | 3D pressure drop analysis .....                     | 5  |
| 5   | Summary .....                                       | 6  |
| 6   | References.....                                     | 7  |
| 7   | Appendix.....                                       | 8  |
| 7.1 | Figures and graphs .....                            | 8  |
| 7.2 | Additional Equations/Formulae .....                 | 15 |

# 1 Introduction

In fluid dynamics, pipe flow is a critical subject and has many applications in the engineering industry and elsewhere, such as chemical processing and HVAC systems.

When designing these pipes, it is imperative to minimise energy losses and ensure efficient fluid flow. However, implementing 90° pipe bends gives rise to certain unwanted flow behaviours, as they introduce complex flow patterns that straight pipe sections do not, such as eddy currents and flow separation [1]. These can greatly increase the energy lost and cause wear on the pipe walls. An indicator of energy loss in the system is a pressure loss across the bend.

This report investigates the flow dynamics in a 90° pipe bend, simulated using both two-dimensional (2D) and three-dimensional (3D) models with STARCCM++. It will explore the effects of varying Reynolds numbers on the behaviour of the fluid, with a primary focus on the effect of a 90-degree pipe bend on the pressure drop within the pipe, which is influenced by factors such as flow velocity, bend radius and pipe diameter. By comparing our simulated pressure drops with theoretical values from empirical models, we will be assessing the accuracy of computational fluid dynamics (CFD) simulations in modelling complex pipe flow.

As the fully developed flow reaches the bend of the pipe, due to centrifugal forces, the velocity and radial pressure at the walls is zero due as described by the following equation:

$$-\frac{\partial P}{\partial r} = \frac{\rho V^2}{r} (1)$$

To account for high pressure along the outer radius and low pressure on the inner radius, the fluid must accelerate on the inside of the bend, while flow deceleration and possible separation occur on the outer wall of the bend.

To determine the pressure drop across the straight sections of pipe (more specifically the inlet and outlet), the Darcy-Weisbach equation [8] can be employed:

$$\Delta P_{Loss} = f \frac{L}{D} \frac{\rho V^2}{2} (2)$$

This accounts for friction factor,  $f$ , which is dependent on the Reynolds number and relative roughness of the pipe.

Whereas for the pressure drop along the bend section of the pipe, various empirical models were studied through a literature review, from which the two focused on within this paper are Oberg's method [2], and Spedding's method [3].

These empirical methods were mainly obtained through experimentation using wind tunnels to simulate fluid flow through bend sections in a pipe for flow of specific Reynolds numbers, which then formed the basis to find specific trends observed through the plotting of variables on graphs.

The four different Reynolds numbers analysed in this paper are:  $Re = 5e4$ ,  $Re = 1e5$ ,  $Re = 5e5$ , and  $Re = 1e6$ . A higher Reynolds number corresponds to a higher turbulence, so we can expect to find more eddies and vortices due to higher flow fluctuations, thus causing increased energy dissipation.

## 2 2-D Results

### 2.1 How the pipe was modelled

The 2D mesh was modelled to be as balanced as possible, in regard to simulation time and accuracy of numerical solutions. The mesh was designed with smaller control volumes nearer the boundary of the pipe and larger nearer the centre, to ensure more accuracy at the boundary of the model accounting for the viscous effects generated by the walls of the pipe. This was also due to the need to balance for the simulation time, since the smaller the control volume, the more iterations need to be run to generate data, therefore only the most important regions of flow had the smaller control volumes. The CFD was then set up with various parameters, likening the flow to be as realistic as possible when ran, but also optimising the efficiency of the simulation. For this reason, the flow was modelled using segregated flow, which is a SIMPLE discretisation scheme. This works by breaking down the continuous flow into discrete values/sub-problems, and each one is individually solved over the whole pipe, and then combined to give the whole flow solution. This is what produces the overall pressure and velocity contours in the simulation. Since the flow modelled in the pipes was turbulent, a method that would be able to accurately predict the development of the turbulent flow was required.

Therefore, the K-epsilon-turbulence model was used to handle the viscous effects of the turbulent fluctuations, as it is the simplest turbulence model for which only boundary conditions and initial conditions need to be supplied, and the Reynolds averaged Navier-Stokes is used to approximate the time-averaged pressures and velocities to the Navier-Stokes equations, since normally modelling turbulent flow can be very computationally demanding. All these factors produce approximate values for the pressures and velocities throughout the pipe using numerical calculations following the turbulence models specified, whilst keeping computational demand low. Proceeding to the commencement of the simulation, the flow inlet and outlet were specified, with the reference pressure being atmospheric pressure, this is so the pressure difference can be calculated with respect to gauge pressure. All the flow is said to enter the inlet, so external fluid flow factors can be deemed negligible, and the flow at the inlet is also modelled to be fully developed as it enters. This is done, as allowing the flow to fully develop within the pipe would require an inlet length of 20-50 diameters, which would be too computationally demanding, and in our case not necessary, however it would likely be required when replicating the experiment in a wind tunnel, or a similar lab setting. The pipe walls are modelled as a boundary, making it so that the fluid must flow through the entire length of the pipe, meaning no flow can permeate through the pipe walls, nor can any external effects disturb the fluid flow within the pipe. Furthermore, when the flow profile is set, the inlet is defined as having high pressure, and the outlet having lower pressure, hence the fluid flows from the inlet towards the outlet. The fluid flow is deemed incompressible, since the fluid speed is negligible compared to the speed of sound, i.e. the speed is less than 0.3 Mach number ( $Ma < 0.3$ ). This does not seem to hold when a Re number of  $1e6$  is used, as its velocity has a value of around  $Ma = 0.45$ , hence its simulated values can be expected to differ from real experimental data. The density and dynamic viscosity of the fluid is modelled as constant, this is due to the fluid being incompressible.

## **2.2 2-D CFD velocity and pressure fields analysis**

The flow in the pipes can be observed and compared as it develops, as all three simulations have a fluid flowing through them with a Reynolds number that corresponds to turbulent flow. Initially pressure drop fluctuates due to the large amount of turbulence experienced by this fluid, eventually stabilising slightly before becoming inconsistent once again. Observing the flow through the bend, a steep pressure gradient is seen around the pipe bend, this is evident in Fig. (1), (3), and (5), where a low-pressure area is created along the wall of the inner bend and a high-pressure zone at the wall of the outer bend radius. In Fig. (2), (4), and (6), the fluid can be seen to lose significant velocity on the outer bend radius due to the high pressure along that section, and the fluid gains speed along the inner radius, accelerating the fluid around the inner bend. This corroborates with Bernoulli's equation, as to conserve it, the fluid would have to gain speed within an area of low pressure and reduce in speed in an area of high pressure, thus it causes the inner bend fluid velocity to be greater than the outer bend velocity in the pipe bend, though this can only be used to describe flow away from the wall, not acted on by shear stresses due to viscosity. A closer visual of the fluid velocity (for the Re number of  $1e5$ ) around the bend can be seen in Figure 8. The high pressure on the outer bend radius can be explained due to fluid impaction, whereas at the inner wall, due to the separation of the boundary layer and formation of a secondary circulation, it results in a low pressure, creating a pressure gradient radially through the bend. Due to the increased velocity of the fluid at the inner bend radius, the fluid at that position experiences greater viscous shear stresses, leading to large energy losses, this causes the fluid to exit the bend with low energy, and since the pressure gradient slowly recovers past the pipe bend, the some of the fluid does not possess enough energy to overcome the increase in pressure, leading to flow separation, and consequently a stagnation point. Although not as evident as in the 3D simulations, this phenomena can be observed in Fig. (2), (4), and (6) after the pipe bend along the inner wall, where the fluid with lower energy creates the blue region of zero velocity at the inner wall, and the fluid with greater energy gets shifted to the centre of the pipe flow. The effects are not as drastic on the fluid at the outer bend radius, since the viscous shear forces do not act as severely on the fluid at that point, due to the viscous shear force being proportional to fluid velocity. After the bend section, the pipe becomes straight again, this leads to the velocity with enough energy to shift to the centre and the area of the average velocity increasing along the pipe cross section, showing the flow to be redeveloping. These effects on the velocity cause the pressure gradient to recover, however due to minor head losses around the pipe bend, the overall effect leads to a pressure loss between the pipe inlet and outlet. The main disparities between the 3 Re number flows were the maximum and minimum pressures and velocities observed as the flow developed. However, this can be evidenced through analysis of the Reynold's number

formula, where the density, viscosity, and diameter values are constant between all three pipes, so as the Reynold's number increases, the average velocity of the fluid also increases, and with that the pressures change accordingly.

## 2.3 Remarks on graphs of results

Generally, for more turbulent flows, energy transfers are maximized, which increases both the range of velocities and pressures of the fluid in flow. This can be seen in the scales of Fig. (2), (4), and (6), as the range of the velocity magnitude for the fluid of  $Re = 5e5$  is significantly larger than that of the fluids with and  $Re$  number of  $1e5$ , and  $5e4$ . Similarly, the range of pressures is also large, as seen in Fig. (1), (3), and (5). This difference can be attributed to the irregular nature of turbulent flow. Unlike laminar flow, velocities go to high and low extremes rather than remaining uniform. Eddies and vortices produced by the flow enhance mixing in fluid, which results in pressure variations, as localized drag forces act within the fluid and interact with the pipe walls, which will be further examined in the following section. The random and chaotic motion of fluid in turbulent flow causes energy dissipation to be more likely to occur after work is done against frictional forces. The CFDs were run until the pressure drop reached a steady state, indicated by the asymptotic behaviour exhibited by Figure 10. This is not reflected within the graphs of residuals (Figure 11) showing the convergence of continuity and x-y momentum, as it would be unnecessary for these to reach steady state, since STARCCM++ has double precision ( $10^{-14}$ ), and values at this order can be considered to be 'noise'. Therefore, the 2D simulations can be seen to converge at 698 iterations for  $Re = 5e4$ , 782 for  $Re = 1e5$ , and 1057 for  $Re = 5e5$ . This can give further insight into what increased turbulence leads to, for the case of the simulations, it causes the software to run more iterations before a steady state is determined.

## 3 3-D Results

Before reaching the bend, specifically near the pipe inlet, the flow exhibited typical turbulent characteristics. As the flow reached the bend, distinct pressure changes were observed aligning with the 2D observations. For 3-D models of fluid flow in a pipe with the same dimensions and geometry as in the 2-D cases, it was seen that fluids with Reynolds' numbers greater than 4000 (fully turbulent flows) behaved similarly to those discussed above: the radially increasing pressure gradient due to frictional forces causes a radially decreasing velocity from the inner to outer curve. However, it can be seen in velocity and pressure contours for 2-D and 3-D models that the 3-D fields display a significant decrease in velocity and pressure after the fluid passes the bend, which cannot be seen in the 2-D profiles (Fig. 12 & 13). This is because the 3-D CFD models account for the eddies and vortices that form after encountering the bend. Secondary flows and boundary layer separation, which result in major head losses (energy losses), are accounted for in the 3-D version, thus pressure and velocity decrease after experiencing shear force at the bend.

The centrifugal force from the inertia of the fluid as it goes around the bend results in the overlapping of the path lines. This can be seen in Fig. (18) and Fig. (19), where streamlines from the inner bend cross over towards the centre and those from the centre move towards the inner wall. These counter-rotating vortices reduce the stability of flow exiting the pipe and increase the viscous shear forces that act on the fluid, causing increased frictional losses.

Characteristics of the secondary flow depend on four main factors: radius of curvature, Reynold's number, pipe diameter and bulk velocity. Intuitively it can be recognised that a larger radius of curvature and pipe diameter would lead to a less sharp turn for fluid at the pipe bend, thus resulting in lower levels of separation and secondary flow. If the ratio between these two variables respectively is smaller than 1.5, the vortices formed are harder to predict and flow separation becomes more apparent than if the ratio was greater than 1.5. The latter effect is what occurs in our case study, where  $r_c = 2D$  [7].

When comparing Fig. (20) and (21), it is made apparent that Dean vortices form after the elbow, and the primary velocity around the pipe centre is reduced significantly. This is conveyed with the symmetrical pattern in the planar velocity profiles, where the outer bend is shown to have areas with high velocity particles and the inner bend has lower velocities as the fluid is pushed outwards due to centrifugal forces during the turn (Figure 15).

In Figure 23, the magnitude of secondary flow can be seen to reduce downstream of the bend as the fluid stabilises. This decay in swirl intensity can be quantified by Eq.(A-1) [1], showing the exponentially decreasing function,  $I_s$ , which is swirl intensity determined using the magnitude of velocity tangential to the pipes cross section after the bend.

Recirculation zones can sometimes form after a sharp enough curve in a pipe. The particles closer to the surface are unable to turn enough so separate from the wall. This is due to an adverse pressure gradient, which also causes fluid particles near the surface to lose momentum and reverse in direction. The area with a cycle of fluid circulating backwards, and forwards again after a stagnation point is known as a recirculation zone. The 3D vector velocity fields for the modelled flow do not display this, as the velocity is never negative, thus reverse flow is not shown to occur. This differs from our expectations, as typically for turbulent flows through 90 degrees bends, recirculation occurs (especially for relatively high Reynold's numbers). The prevalent effect of these flow phenomena is an increase in mixing within the fluid. This would increase expected pressure drop and also make the modelling process more complex if it were to account for all these occurrences.

When  $Re = 1e6$ , the flow exhibited a significantly larger pressure drop after passing through the bending region compared to the other two cases. This can be credited to the flow's high velocity which amplifies the effects of friction and turbulent mixing, especially in the bend, leading to greater energy dissipation as the flow navigates the bend. The higher kinetic energy initially present is lost rapidly through friction and turbulence as the flow is forced to change direction. This results in a stronger pressure drop in the bend and post-bend regions. The intensified turbulence at this high Reynolds number would contribute to more chaotic flow patterns, including larger stagnation zones and reversing flows.

## 4 Analysis of pressure drop data

Table 1: Pipe model constants

|                             |                           |
|-----------------------------|---------------------------|
| Inlet length ( $L_{in}$ )   | 0.2 m                     |
| Outlet length ( $L_{out}$ ) | 0.4 m                     |
| Pipe diameter (D)           | 0.1 m                     |
| Pipe radius (R)             | 0.05 m                    |
| Bend radius ( $r_c$ )       | 0.2 m                     |
| Density of fluid ( $\rho$ ) | 1.18415 kg/m <sup>3</sup> |
| Dynamic viscosity ( $\mu$ ) | 1.85508e-5 Pa-s           |

### 4.1 2D pressure drop analysis

**Calculating the average velocity of fluid through the three pipe cases:**

Using the Reynolds number formula:

$$Re = \frac{\rho V D}{\mu} (3)$$

Rearranging for V, it is possible to obtain the average fluid velocity through the pipe.

Table 2: Pressure drop and average velocity associated with the Reynolds number for each modelled 2D case

| Reynolds number | Simulation pressure drop (Pa) | Average fluid velocity (m/s) |
|-----------------|-------------------------------|------------------------------|
| 5e4 (50000)     | 2.861                         | 7.83                         |
| 1e5 (100000)    | 10.03                         | 15.7                         |
| 5e5 (500000)    | 174.4                         | 78.3                         |

**Oberg machinery's handbook method [2]:**

Head loss due to pipe bend:

$$h_l = K_b \frac{V^2}{2g} (4)$$

Where,

$$K_b = f \frac{L_e}{D}$$

The Darcy friction factor can be determined for each Reynolds number using the Moody chart (Figure 24), and the value of  $L_e/D$  can be found using the table seen in Figure 25, which is found to be 12, due to  $r_c/D = 2$ . Therefore, the pressure drop due to the bend is given by:

$$\Delta P_{bend} = \rho g h_l = f \frac{L_e}{D} \frac{\rho V^2}{2} \quad (5)$$

Then the pressure drop due to the inlet and outlet need to be added to the bend pressure drop, which can be calculated with the Darcy-Weisbach equation (Eq. 2):

$$\Delta P_{straight} = f \frac{(L_{in} + L_{out})}{D} \frac{\rho V^2}{2}$$

The total pressure drop can then be calculated as follows:

$$\Delta P_{total} = \Delta P_{bend} + \Delta P_{straight}$$

*Table 3: Calculated pressure drop and percentage variation between empirical calculations and simulation models using Oberg's method*

| Reynolds number | Calculated pressure drop (Pa) | Variation (%) |
|-----------------|-------------------------------|---------------|
| 5e4 (50000)     | 13.41                         | 368.5         |
| 1e5 (100000)    | 47.08                         | 369.4         |
| 5e5 (500000)    | 850.1                         | 387.4         |

#### **Spedding's method [3]:**

The total bend loss coefficient forms a relation with the pressure drop as follows:

$$K_T = \frac{\Delta P_{bend}}{0.5 \rho V^2} \quad (6)$$

Which can be rearranged to determine the pressure loss.

The total bend loss can be determined using Eq. 7, since  $Re(R/r_c)^2 > 91$ , for all Re values used,

$$K_T = 0.00241 \alpha \theta Re^{-0.17} (r_c/R)^{0.84} \quad (7)$$

Where for a bend angle,  $\theta = 90$ , and  $r_c/R < 19.7$ ,

$$\alpha = 0.95 + 17.2 (r_c/R)^{-1.96}$$

Using all these relationships and formulae, the pressure drops for each Re number can be found, as this model includes the pressure loss in the inlet, outlet, and bend of the pipe.

*Table 4: Calculated pressure drop and percentage variation between empirical calculations and simulation models using Spedding's method*

| Reynolds number | Calculated pressure drop (Pa) | Variation (%) |
|-----------------|-------------------------------|---------------|
| 5e4 (50000)     | 7.058                         | 146.7         |
| 1e5 (100000)    | 26.97                         | 168.8         |
| 5e5 (500000)    | 623.7                         | 257.6         |

## **4.2 3D pressure drop analysis**

*Table 5: Pressure drop and average velocity associated with the Reynolds number for each modelled 3D case*

| Reynolds number | Simulation pressure drop (Pa) | Average fluid velocity (m/s) |
|-----------------|-------------------------------|------------------------------|
| 1e5 (100000)    | 34.27                         | 15.7                         |
| 5e5 (500000)    | 723.4                         | 78.3                         |
| 1e6 (1000000)   | 2569                          | 156.7                        |

The following calculations are done using the same respective methodologies for Oberg and Crawford, as shown in the 2D pressure drop analysis section.

#### Oberg's method [2]:

Table 6: Calculated pressure drop and percentage variation between empirical calculations and 3D simulation models using Oberg's method

| Reynolds number | Calculated pressure drop (Pa) | Variation (%) |
|-----------------|-------------------------------|---------------|
| 1e5 (100000)    | 47.08                         | 37.4          |
| 5e5 (500000)    | 850.1                         | 17.5          |
| 1e6 (1000000)   | 3008                          | 17.1          |

#### Spedding's method [3]:

Table 7: Calculated pressure drop and percentage variation between empirical calculations and 3D simulation models using Spedding's method

| Reynolds number | Calculated pressure drop (Pa) | Variation (%) |
|-----------------|-------------------------------|---------------|
| 1e5 (100000)    | 26.97                         | 21.3          |
| 5e5 (500000)    | 623.7                         | 13.8          |
| 1e6 (1000000)   | 2443                          | 4.92          |

#### Analysis:

For the Oberg method, there are large variations between the 2D empirical and simulations values for pressure drop, whereas the 3D pressure drops, show slight variation. This can be attributed to the complexity and realism of the models, since the 2D does not consider flow in the 3<sup>rd</sup> dimension, hence many factors including increased viscous shear stresses acting on the fluid due to Dean vortices, and secondary flow are not accounted for, leading to discrepancies between empirical calculations and simulation values. This can also be explained further, since these empirical models were formed on the basis of a 3D model in mind, therefore it is to be expected that there is a smaller variation in the 3D models. Another thing to note is that there is an increase in variation as the Re number increases, this can be associated with the nature of turbulent flow, as it gets less predictable the more turbulent the flow is.

The Spedding method shows better reliability for the 2D and 3D results, while also producing lower variation for the lower Re numbers in the 2D results. However, for the 3D, it yields a better overall average for variation, and it produces a better set of overall values, which seem to decrease in variation as the Re number becomes greater, contrary to the Oberg method. This can be attributed to the fact that Oberg's method uses an approximation for  $L_e/D$  based of the ratio of bend radius and diameter, therefore leading to greater inconsistency, whereas Spedding's method utilizes a more complete empirical model, which also includes the head loss in the straight sections of pipe in its model.

## 5 Summary

This report focuses on the usage of STARCCM as a CFD tool, utilizing it to analyse the pressure drop in a pipe section including a 90-degree bend through numerically solving over the mesh of the pipe. These results were then to be compared against existing empirical methods, the two of which this paper focuses on being Oberg's method and Spedding's method. From this comparison, a few key observations could be made, with the most prominent being that the Spedding method showed greater reliability across both the 2D and 3D models, with an average variation of 191% for the 2D case, and an average variation of 13.3% for the 3D case. This is a vast improvement against the Oberg empirical model, which gave an average variation of 375% for 2D, and an average variation of 24% for the 3D case. Upon closer inspection, it is evident that the 3D models provided a nearer approximation to the values of pressure drop generated by the simulation, this can be explained through fact that the 3D mesh and model were much more complex, providing a closer solution to realistic expectations. From sections 2 and 3, the flow phenomena were further explained through the analysis of the flow profiles generated. The main points discussed in section 2, were that the majority of the pressure drop was generated due to the bend, being a result of the pressure gradient generated radially along the bend radius, causing the



fluid to accelerate along the inner bend, and slow along the outer bend. This increased velocity at the inner bend caused a proportionally significant increase in viscous shear stress opposing the fluid, hence increasing frictional losses. The three flow profiles did not exhibit substantial differences in the 2D case.

Section 3 expanded on the complexity of the flow profiles, showing a more complex fluid motion through the pipe section. Though the points made in section 2 still held here, there were additional motions visible, some of the most prominent being the appearance of Dean vortices after the 90-degree bend. This vortical motion added to the pressure losses over the pipe section, leading to differing pressure drop values, compared to the 2D case, being recorded over the Re numbers tested. An additional phenomenon encountered was flow separation, which was exhibited in the contour diagrams for each flow profile after the pipe bend, nearer the inner bend radius. This led to a streamline with a different velocity being created, which lead to flow being unable to cross this separation streamline.

To conclude, it can be seen that Spedding's method provides a closer approximation to expected pressure drop values, therefore if the pressure drop was needed to be determined in a pipe system, I would propose to use Spedding's method. This is due to the fact that Oberg's method includes approximations and does not include the pressure drop due to straight sections, whereas Spedding's model incorporates both the bend losses and major losses due to straight pipe sections.

## 6 References

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## 7 Appendix

### 7.1 Figures and graphs

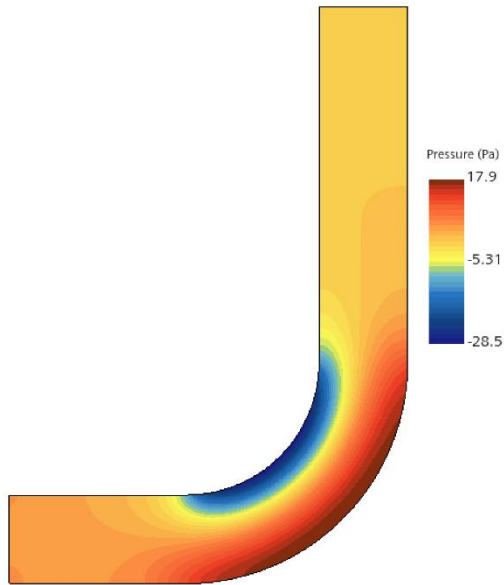


Figure 1: Pressure contours within 90-degree pipe bend for fluid with Reynold's number of  $5e4$

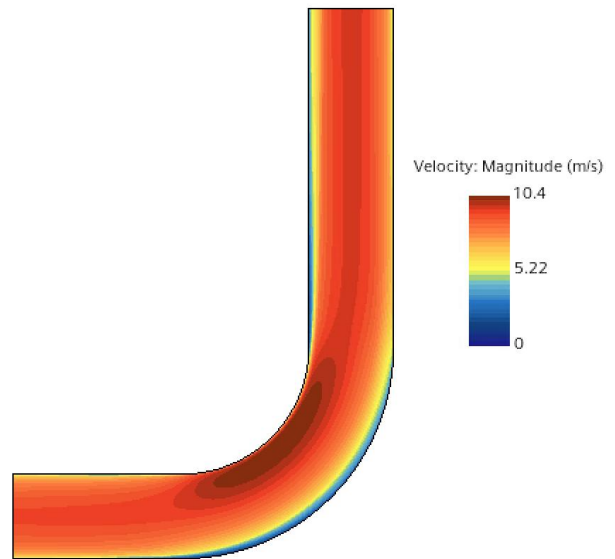


Figure 2: Velocity contours within 90-degree pipe bend for fluid with Reynold's number of  $5e4$

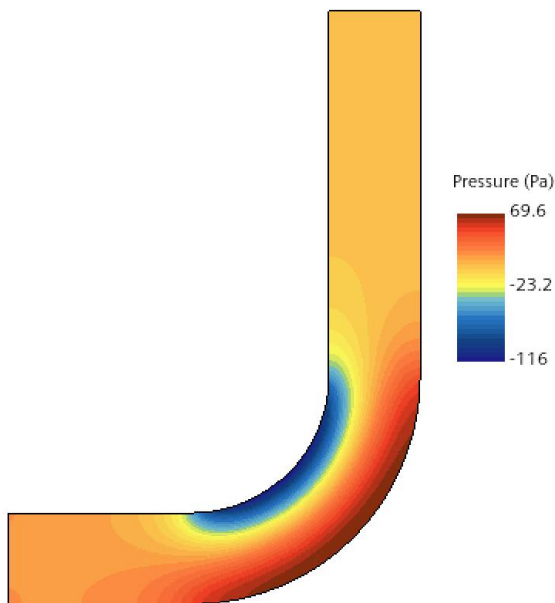


Figure 3: Pressure contours within 90-degree pipe bend for fluid with Reynold's number of  $1e5$

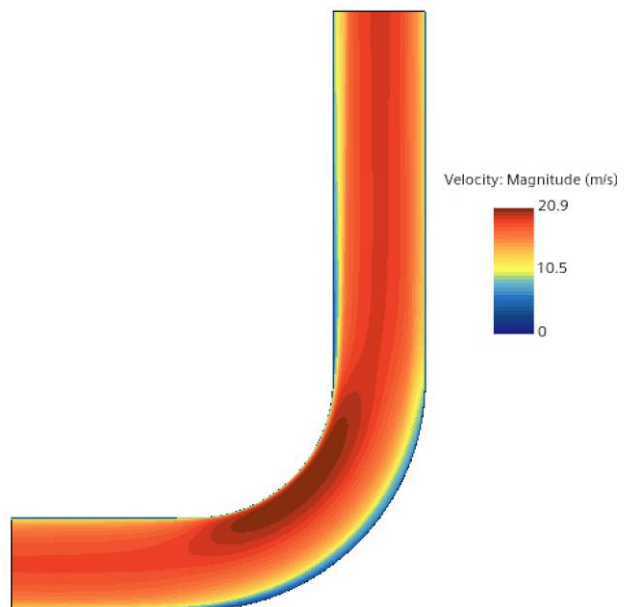


Figure 4: Velocity contours within 90-degree pipe bend for fluid with Reynold's number of  $1e5$

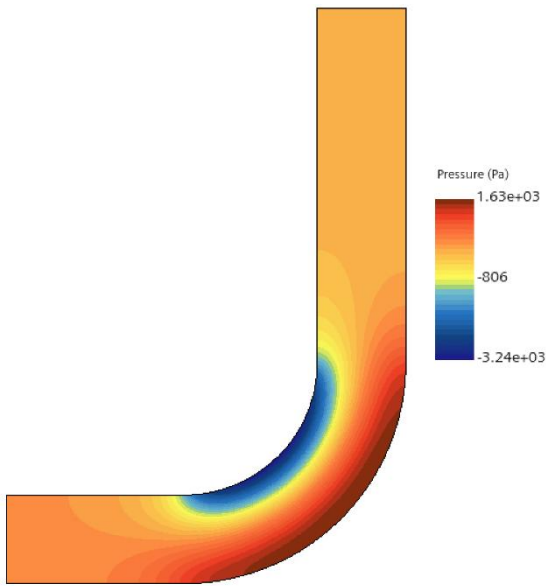


Figure 5: Pressure contours within 90-degree pipe bend for fluid with Reynold's number of  $5e5$

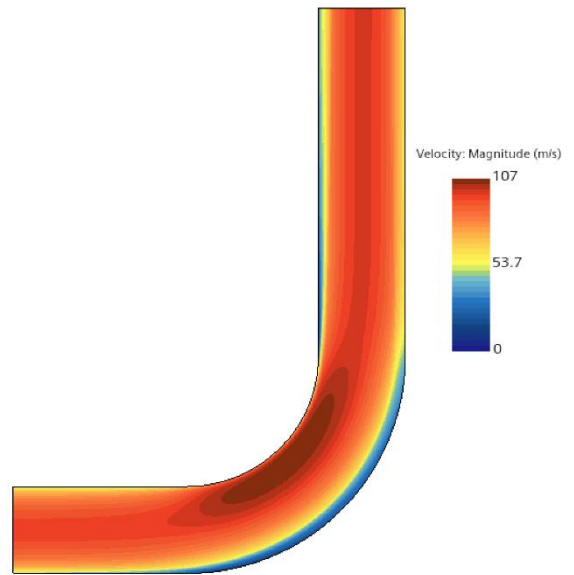


Figure 6: Velocity contours within 90-degree pipe bend for fluid with Reynold's number of  $5e5$

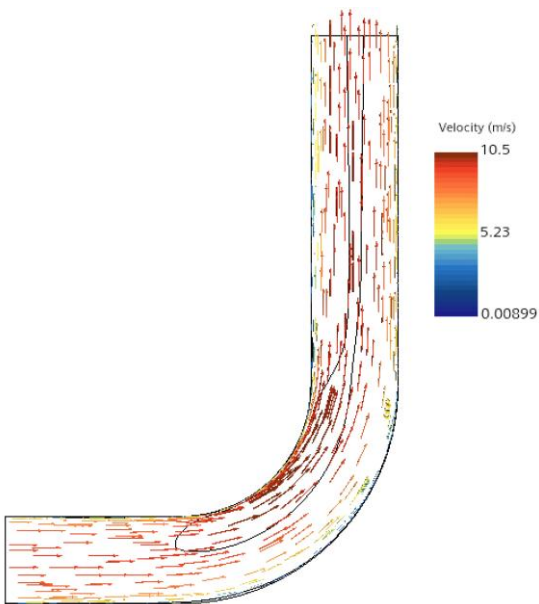


Figure 7: Velocity vector representation of pipe flow for fluid with Reynold's number of  $5e4$

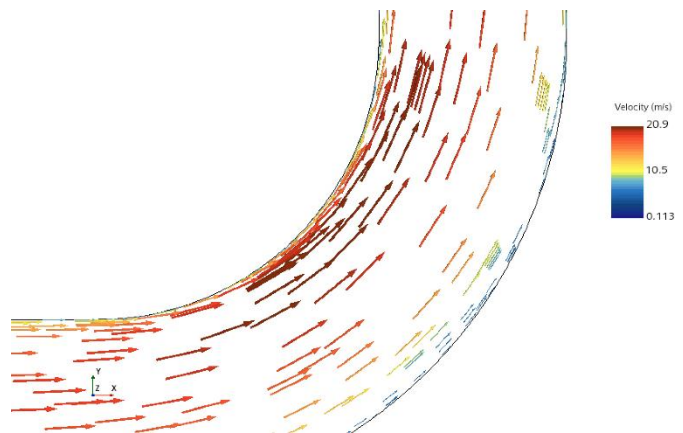


Figure 8: Velocity vector representation of pipe flow within 90-degree bend for fluid with Reynold's number of  $1e5$

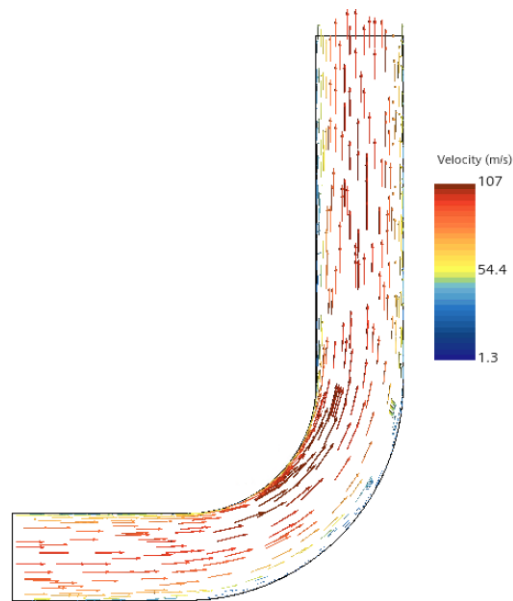


Figure 9: Velocity vector representation of pipe flow for fluid with Reynold's number of  $5e5$

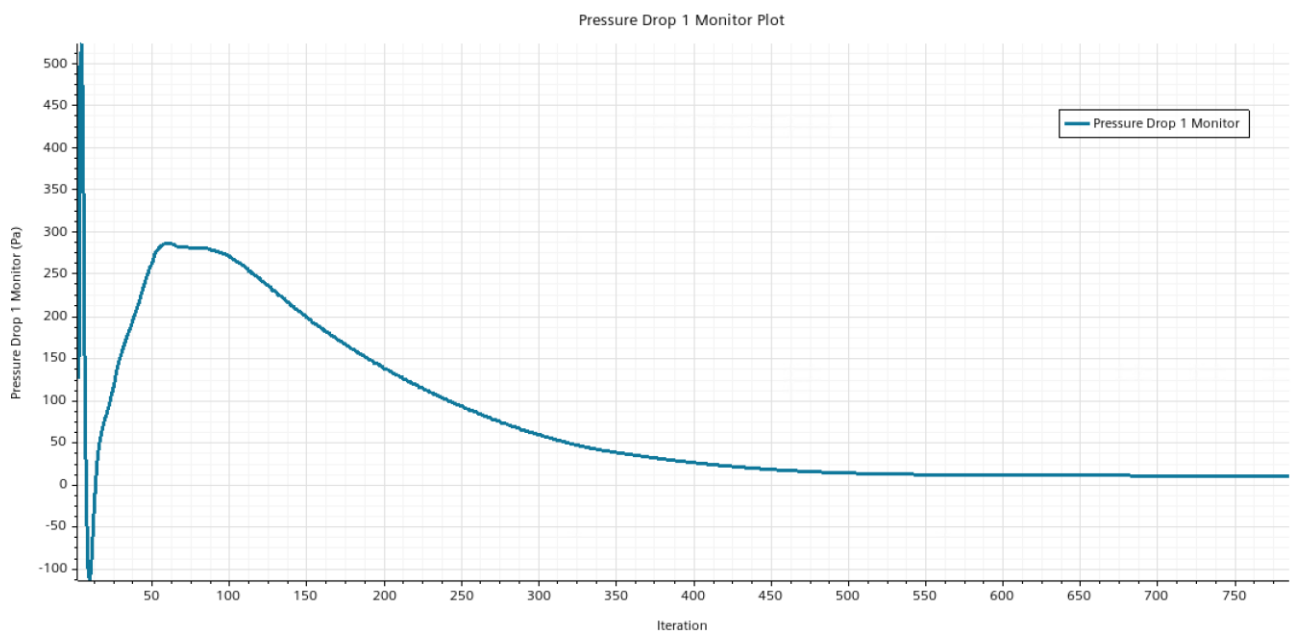


Figure 10: Pressure drop against iterations plot for  $Re = 1e5$

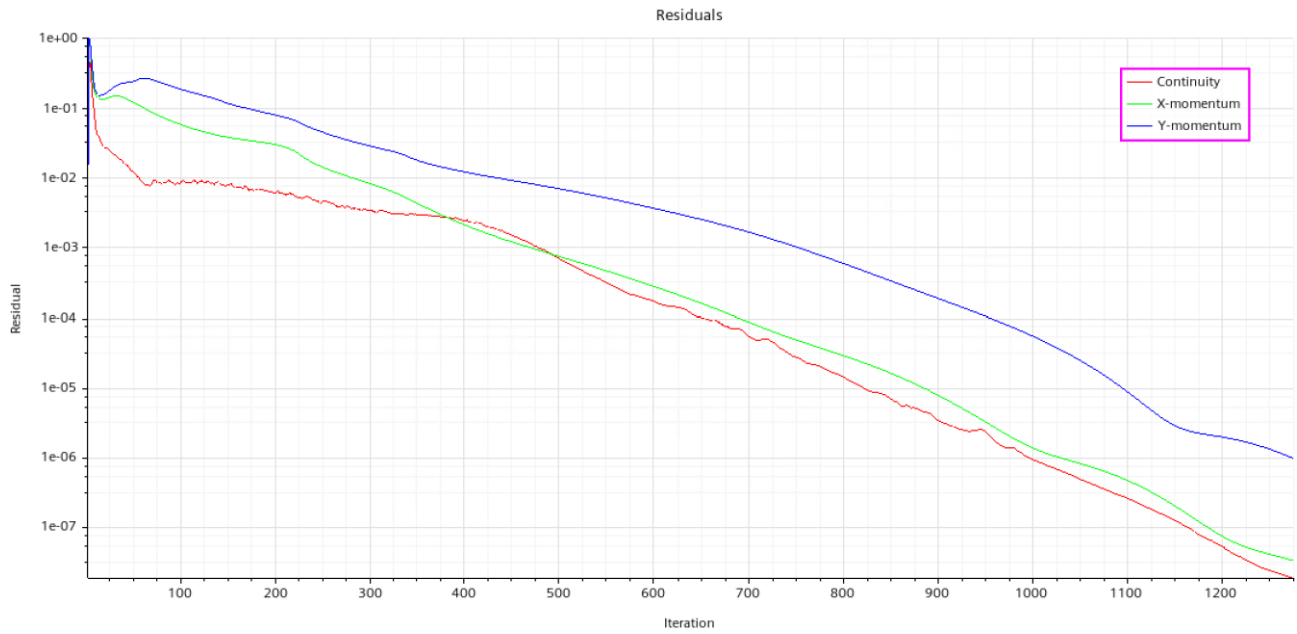


Figure 11: Plot of residuals for  $Re = 1e5$

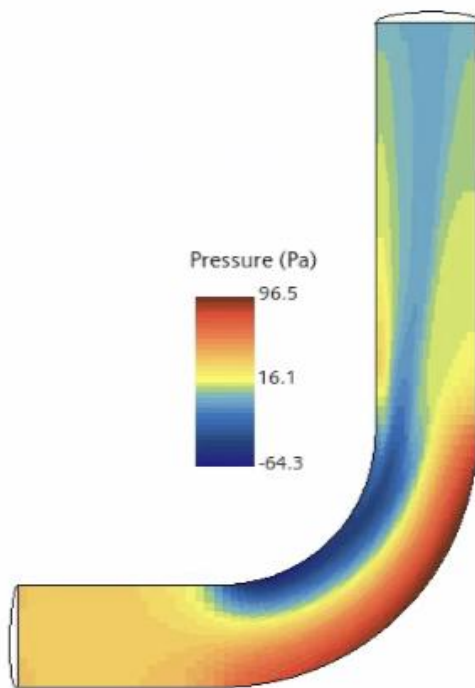


Figure 12: Pressure contours within 90-degree pipe bend for fluid with Reynold's number of  $1e5$  (3D model)

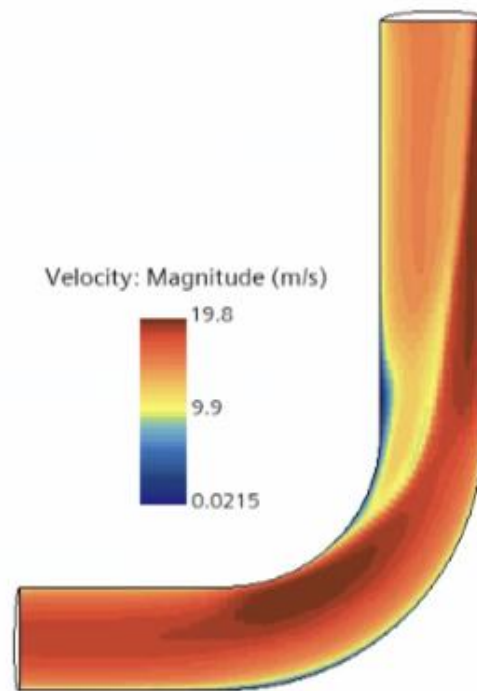


Figure 13: Velocity contours within 90-degree pipe bend for fluid with Reynold's number of  $1e5$  (3D model)

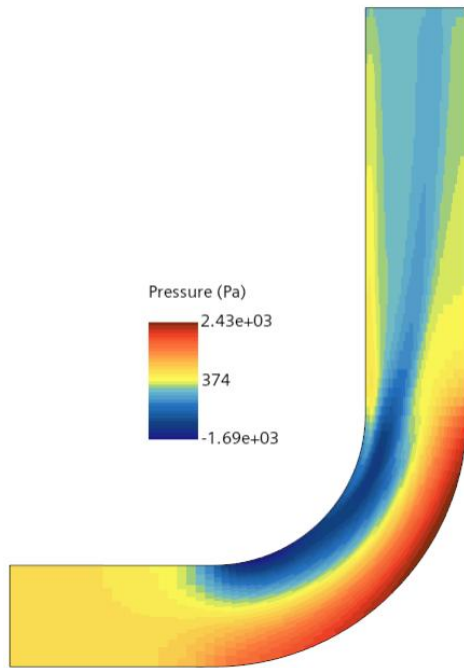


Figure 14: Pressure contours within 90-degree pipe bend for fluid with Reynold's number of  $5e5$  (3D model)

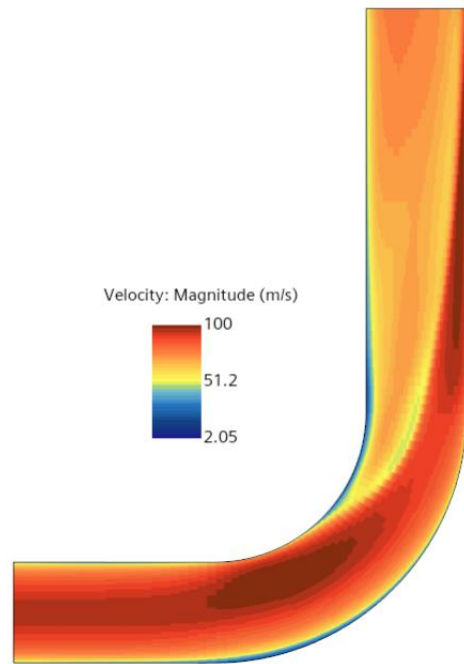


Figure 15: Velocity contours within 90-degree pipe bend for fluid with Reynold's number of  $5e5$  (3D model)

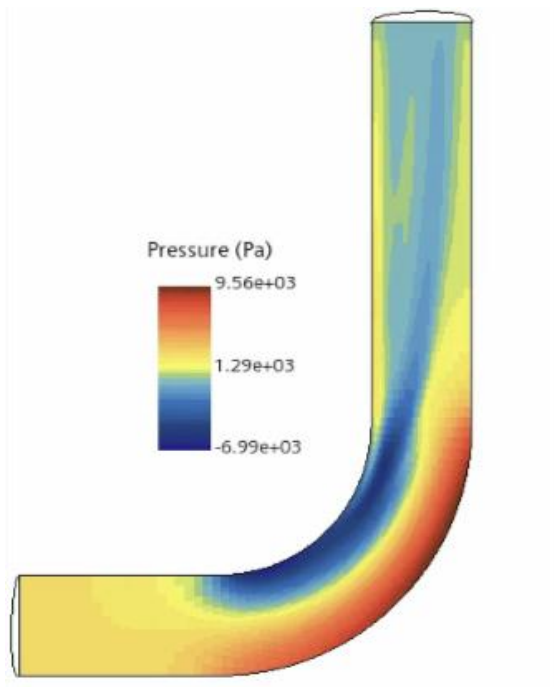


Figure 16: Pressure contours within 90-degree pipe bend for fluid with Reynold's number of  $1e6$  (3D model)

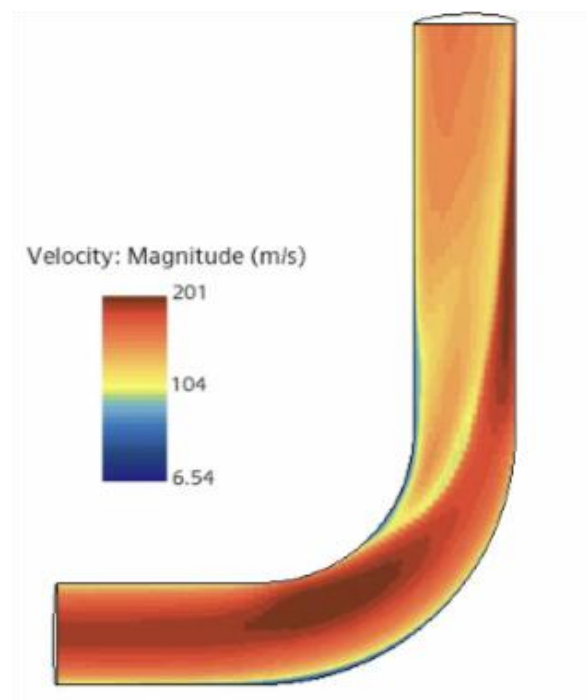


Figure 17: Velocity contours within 90-degree pipe bend for fluid with Reynold's number of  $1e6$  (3D model)



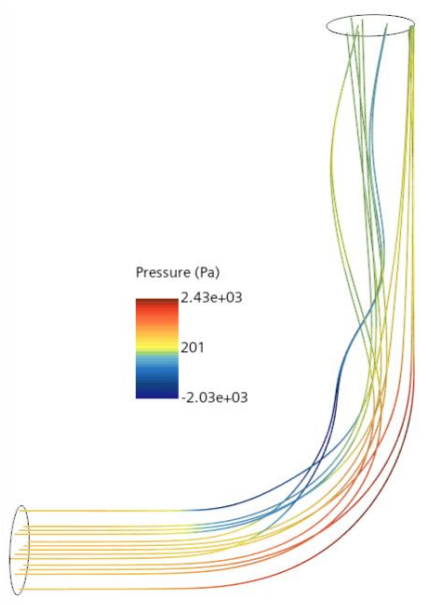


Figure 18: Pressure profile streamlines for  $Re = 5e5$  (3D model)

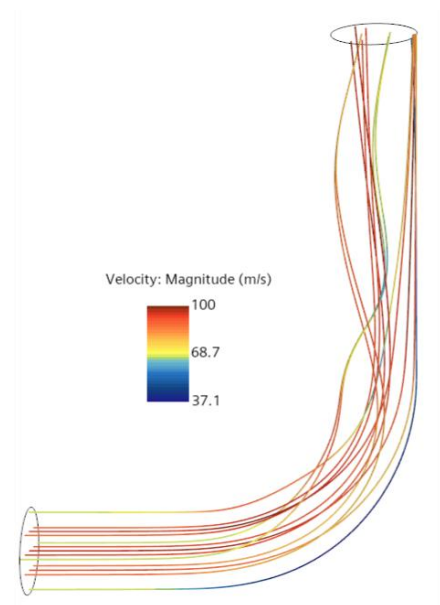


Figure 19: Velocity profile streamlines for  $Re = 5e5$  (3D model)

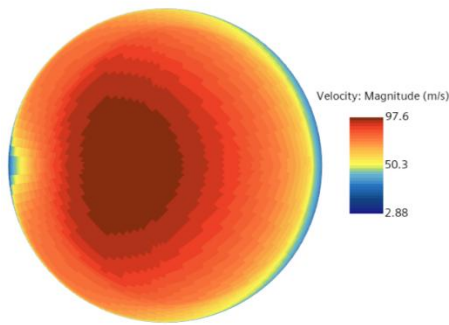


Figure 20: Velocity contours through section of pipe before bend (3D model)

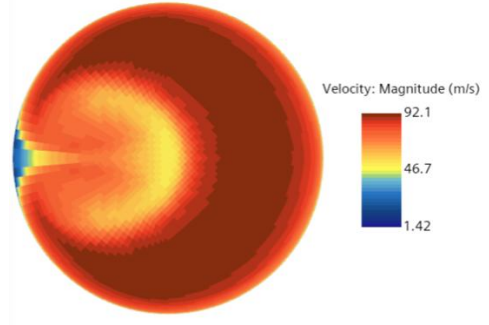


Figure 21: Velocity contours through section of pipe after bend (3D model)

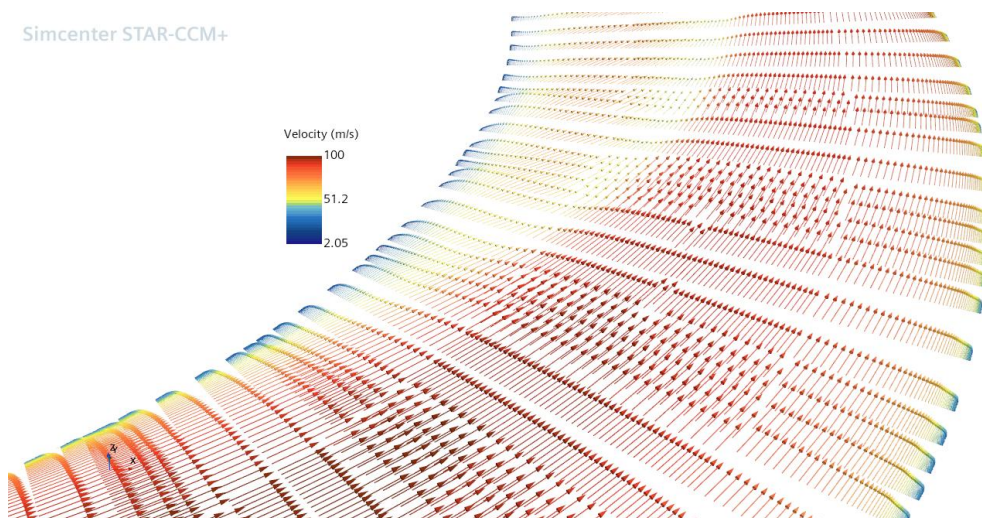


Figure 22: Velocity magnitude in vectorial representation at pipe bend (3D model)

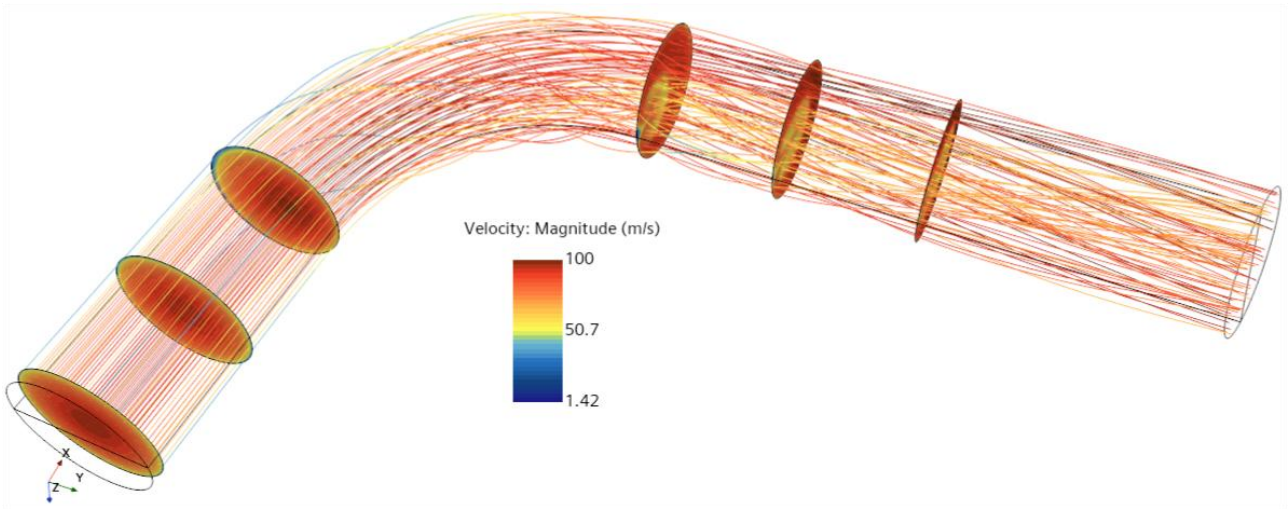


Figure 23: Velocity magnitudes shown for different sections and streamlines for  $Re = 5e5$  (3D model)

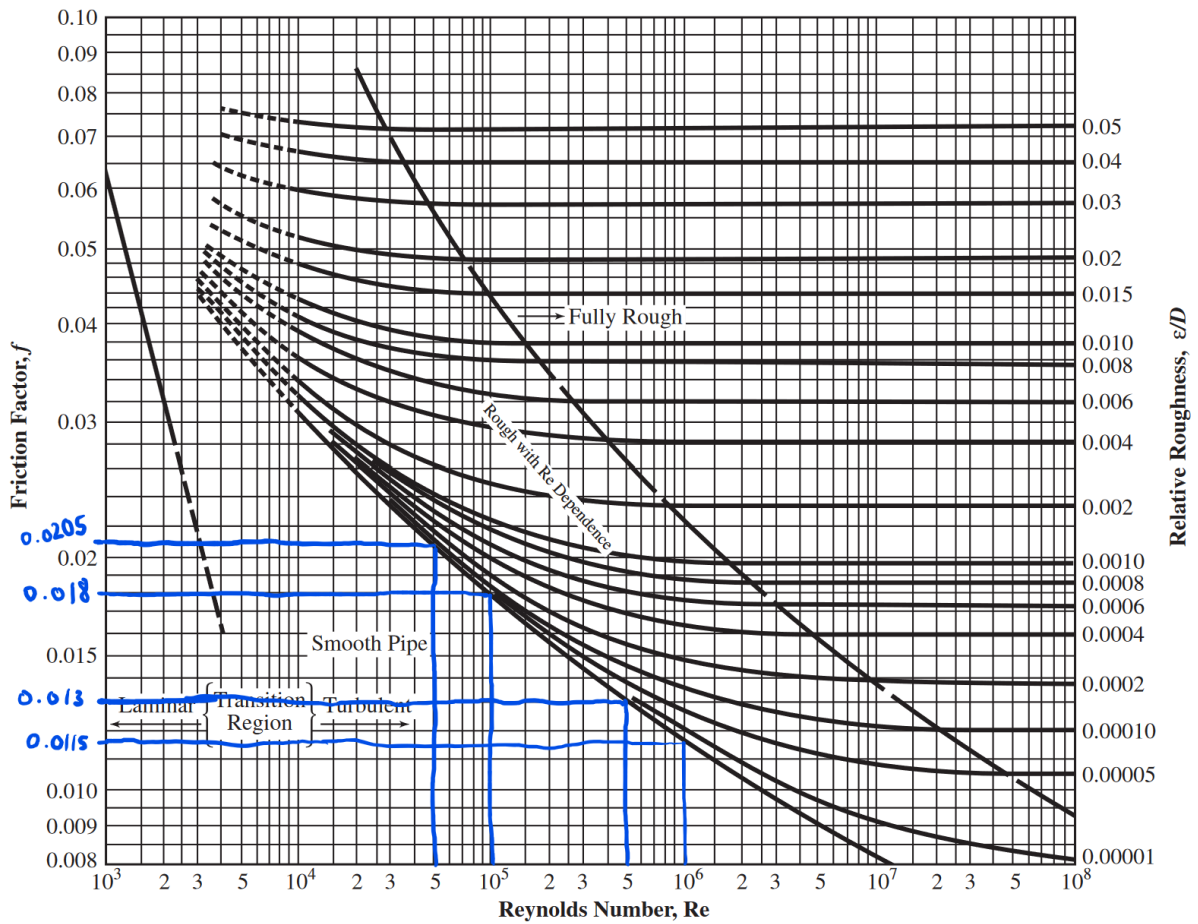


Figure 24: Moody chart showing relevant friction factors for Reynolds numbers investigated



**Table 25. Bend Equivalent Length Ratio,  $L_e/D$** 

| Bend Type   | Bend Angle | $R/D$ | $L_e/D$ | Bend Type         | Bend Angle | $R/D$ | $L_e/D$ |
|-------------|------------|-------|---------|-------------------|------------|-------|---------|
| Radius Bend | 90°        | 1     | 20      | Miter Bend        | 30°        | ...   | 8       |
|             |            | 2     | 12      |                   | 45°        | ...   | 15      |
|             |            | 3     | 12      |                   | 60°        | ...   | 25      |
|             |            | 4     | 14      |                   | 90°        | ...   | 60      |
|             |            | 6     | 17      | Close Return Bend | 180°       | ...   | 50      |
|             |            | 8     | 24      |                   |            |       |         |
|             |            | 10    | 30      |                   |            |       |         |
|             |            | 12    | 34      |                   |            |       |         |
|             |            | 16    | 42      |                   |            |       |         |
|             |            |       |         |                   |            |       |         |

$R/D$  is the bend mean radius divided by pipe diameter

Figure 25: Table from Oberg machinery handbook [2]

## 7.2 Additional Equations/Formulae

$$\frac{I_s}{I_{s0}} = \exp\left(-\beta_s \frac{L}{D}\right)(A - 1)$$

Where:  $\beta_s$  represents the rate at which the secondary flow intensity decreases downstream of the elbow,  $I_s$  is the swirl intensity,  $I_{s0}$  is the initial swirl intensity at the bend,  $L/D$  is the length travelled downstream of the bend over the pipe diameter.